

Report - Bayesian Subset Selection

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Data

We used the Chapman Data, which contains the following variables:

- Ag: age of in years;
- S: systolic blood pressure in millimeters of mercury;
- D: diastolic blood pressure in millimeters of mercury;
- Ch: cholesterol in milligrams per DL;
- H: height in inches;
- W: weight in pounds; CNT: 1 if a coronary incident had occurred in the previous ten years, and 0 otherwise.

```
head(data)
```

```
## # A tibble: 6 x 7
##   Ag      S      D      Ch      H      W      CNT
##   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1    44   124    80   254    70   190     0
## 2    35   110    70   240    73   216     0
## 3    41   114    80   279    68   178     0
## 4    31   100    80   284    68   149     0
## 5    61   190   110   315    68   182     1
## 6    61   130    88   250    70   185     0
```

Firstly, we created a list of formulas to be used in the Bayesian subset selection:

```
head(formulas)
```

```
## [[1]]
## CNT ~ Ag
## <environment: 0x0000026ee990dd88>
##
## [[2]]
## CNT ~ S
## <environment: 0x0000026ee990e0d0>
##
## [[3]]
## CNT ~ D
## <environment: 0x0000026ee990e338>
##
```

```
## [[4]]
## CNT ~ Ch
## <environment: 0x0000026ee990e5a0>
##
## [[5]]
## CNT ~ H
## <environment: 0x0000026ee990e878>
##
## [[6]]
## CNT ~ W
## <environment: 0x0000026ee990eb50>
```

For each formula, we fit a Bayesian logistic regression model using the `glm.post` function from the `hdbayes` package. We used 2000 iterations for warmup and 2000 iterations for sampling.

```
# run hdbayes
fits_bayes <- lapply(formulas, function(x)
  glm.post(x,
    family = binomial(link = "logit"),
    data = list(data),
    parallel_chains = ncores,
    iter_warmup = 2000,
    iter_sampling = 2000,
    refresh = 0,
  ))
```

```
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 0.4 seconds.
## Chain 2 finished in 0.4 seconds.
## Chain 3 finished in 0.4 seconds.
## Chain 4 finished in 0.4 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 0.4 seconds.
## Total execution time: 0.6 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 3 finished in 0.6 seconds.
## Chain 1 finished in 0.8 seconds.
## Chain 2 finished in 0.7 seconds.
## Chain 4 finished in 0.7 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 0.7 seconds.
## Total execution time: 0.9 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 0.8 seconds.
## Chain 2 finished in 0.8 seconds.
## Chain 3 finished in 0.8 seconds.
## Chain 4 finished in 0.8 seconds.
```

```

##
## All 4 chains finished successfully.
## Mean chain execution time: 0.8 seconds.
## Total execution time: 1.0 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 0.6 seconds.
## Chain 2 finished in 0.6 seconds.
## Chain 3 finished in 0.6 seconds.
## Chain 4 finished in 0.7 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 0.6 seconds.
## Total execution time: 0.9 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 1.4 seconds.
## Chain 2 finished in 1.4 seconds.
## Chain 3 finished in 1.4 seconds.
## Chain 4 finished in 1.4 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 1.4 seconds.
## Total execution time: 1.6 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 0.7 seconds.
## Chain 2 finished in 0.7 seconds.
## Chain 3 finished in 0.7 seconds.
## Chain 4 finished in 0.8 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 0.7 seconds.
## Total execution time: 1.0 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 4 finished in 1.0 seconds.
## Chain 1 finished in 1.1 seconds.
## Chain 2 finished in 1.1 seconds.
## Chain 3 finished in 1.1 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 1.1 seconds.
## Total execution time: 1.3 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 1.2 seconds.
## Chain 2 finished in 1.2 seconds.
## Chain 3 finished in 1.1 seconds.

```

```

## Chain 4 finished in 1.2 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 1.2 seconds.
## Total execution time: 1.5 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 0.9 seconds.
## Chain 3 finished in 0.8 seconds.
## Chain 2 finished in 1.0 seconds.
## Chain 4 finished in 1.0 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 0.9 seconds.
## Total execution time: 1.3 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 2.5 seconds.
## Chain 2 finished in 2.4 seconds.
## Chain 3 finished in 2.5 seconds.
## Chain 4 finished in 2.8 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 2.6 seconds.
## Total execution time: 3.0 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 1.0 seconds.
## Chain 2 finished in 1.1 seconds.
## Chain 3 finished in 1.0 seconds.
## Chain 4 finished in 1.0 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 1.0 seconds.
## Total execution time: 1.4 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 1.7 seconds.
## Chain 2 finished in 1.6 seconds.
## Chain 3 finished in 1.7 seconds.
## Chain 4 finished in 1.8 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 1.7 seconds.
## Total execution time: 2.0 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 1.0 seconds.
## Chain 2 finished in 0.9 seconds.

```

```

## Chain 3 finished in 1.0 seconds.
## Chain 4 finished in 1.0 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 1.0 seconds.
## Total execution time: 1.2 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 4 finished in 2.4 seconds.
## Chain 1 finished in 2.6 seconds.
## Chain 3 finished in 2.5 seconds.
## Chain 2 finished in 2.7 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 2.5 seconds.
## Total execution time: 2.9 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 1.2 seconds.
## Chain 3 finished in 1.1 seconds.
## Chain 4 finished in 1.0 seconds.
## Chain 2 finished in 1.2 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 1.1 seconds.
## Total execution time: 1.4 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 3 finished in 1.1 seconds.
## Chain 4 finished in 1.0 seconds.
## Chain 1 finished in 1.3 seconds.
## Chain 2 finished in 1.2 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 1.2 seconds.
## Total execution time: 1.4 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 2 finished in 2.6 seconds.
## Chain 3 finished in 2.6 seconds.
## Chain 4 finished in 2.6 seconds.
## Chain 1 finished in 2.8 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 2.6 seconds.
## Total execution time: 3.0 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 3 finished in 1.1 seconds.

```

```

## Chain 1 finished in 1.3 seconds.
## Chain 4 finished in 1.2 seconds.
## Chain 2 finished in 1.4 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 1.3 seconds.
## Total execution time: 1.6 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 2.3 seconds.
## Chain 2 finished in 2.2 seconds.
## Chain 4 finished in 2.2 seconds.
## Chain 3 finished in 2.3 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 2.3 seconds.
## Total execution time: 2.6 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 1.0 seconds.
## Chain 2 finished in 1.1 seconds.
## Chain 3 finished in 1.0 seconds.
## Chain 4 finished in 1.0 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 1.0 seconds.
## Total execution time: 1.2 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 2.4 seconds.
## Chain 2 finished in 2.4 seconds.

## Warning: Chain 3 finished unexpectedly!

## Chain 3 finished in 2.5 seconds.
## Chain 4 finished in 2.5 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 2.5 seconds.
## Total execution time: 2.7 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 2 finished in 2.1 seconds.
## Chain 4 finished in 2.1 seconds.
## Chain 1 finished in 2.3 seconds.
## Chain 3 finished in 2.1 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 2.2 seconds.
## Total execution time: 2.4 seconds.

```

```

##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 1.4 seconds.
## Chain 3 finished in 1.4 seconds.
## Chain 4 finished in 1.4 seconds.
## Chain 2 finished in 1.6 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 1.4 seconds.
## Total execution time: 1.7 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 2 finished in 3.2 seconds.
## Chain 3 finished in 3.2 seconds.
## Chain 4 finished in 3.3 seconds.
## Chain 1 finished in 3.5 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 3.3 seconds.
## Total execution time: 3.7 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 1.6 seconds.
## Chain 2 finished in 1.6 seconds.
## Chain 3 finished in 1.5 seconds.
## Chain 4 finished in 1.6 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 1.5 seconds.
## Total execution time: 3.7 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 1.6 seconds.
## Chain 2 finished in 1.6 seconds.
## Chain 3 finished in 1.6 seconds.
## Chain 4 finished in 1.5 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 1.6 seconds.
## Total execution time: 1.8 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 3.3 seconds.
## Chain 3 finished in 3.2 seconds.
## Chain 2 finished in 3.4 seconds.
## Chain 4 finished in 3.4 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 3.3 seconds.

```

```

## Total execution time: 3.7 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 1.5 seconds.
## Chain 4 finished in 1.5 seconds.
## Chain 2 finished in 1.5 seconds.
## Chain 3 finished in 1.5 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 1.5 seconds.
## Total execution time: 1.8 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 3 finished in 3.1 seconds.
## Chain 4 finished in 3.1 seconds.
## Chain 1 finished in 3.3 seconds.
## Chain 2 finished in 3.3 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 3.2 seconds.
## Total execution time: 3.5 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 2 finished in 1.4 seconds.
## Chain 3 finished in 1.4 seconds.
## Chain 4 finished in 1.5 seconds.
## Chain 1 finished in 1.6 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 1.5 seconds.
## Total execution time: 1.8 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 2 finished in 3.5 seconds.
## Chain 3 finished in 3.5 seconds.
## Chain 4 finished in 3.4 seconds.
## Chain 1 finished in 3.7 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 3.5 seconds.
## Total execution time: 3.8 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 2 finished in 2.0 seconds.
## Chain 4 finished in 1.9 seconds.
## Chain 3 finished in 2.1 seconds.
## Chain 1 finished in 2.2 seconds.
##
## All 4 chains finished successfully.

```



```

## Mean chain execution time: 2.0 seconds.
## Total execution time: 2.4 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 3.6 seconds.
## Chain 2 finished in 3.5 seconds.
## Chain 4 finished in 3.6 seconds.
## Chain 3 finished in 3.8 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 3.6 seconds.
## Total execution time: 3.9 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 2 finished in 2.2 seconds.
## Chain 3 finished in 2.1 seconds.
## Chain 1 finished in 2.3 seconds.
## Chain 4 finished in 2.3 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 2.2 seconds.
## Total execution time: 2.5 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 4 finished in 3.1 seconds.
## Chain 1 finished in 3.4 seconds.
## Chain 2 finished in 3.3 seconds.
## Chain 3 finished in 3.3 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 3.3 seconds.
## Total execution time: 3.6 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 1.5 seconds.
## Chain 2 finished in 1.5 seconds.
## Chain 4 finished in 1.6 seconds.
## Chain 3 finished in 1.8 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 1.6 seconds.
## Total execution time: 2.1 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 2 finished in 3.3 seconds.
## Chain 3 finished in 3.4 seconds.
## Chain 4 finished in 3.4 seconds.
## Chain 1 finished in 3.5 seconds.
##

```

```

## All 4 chains finished successfully.
## Mean chain execution time: 3.4 seconds.
## Total execution time: 3.7 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 3.1 seconds.
## Chain 2 finished in 3.1 seconds.
## Chain 4 finished in 3.2 seconds.
## Chain 3 finished in 3.4 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 3.2 seconds.
## Total execution time: 3.7 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 1.6 seconds.
## Chain 2 finished in 1.6 seconds.
## Chain 3 finished in 1.6 seconds.

## Warning: Chain 4 finished unexpectedly!

## Warning: 1 chain(s) finished unexpectedly!

## The remaining chains had a mean execution time of 1.9 seconds.

## Warning: The returned fit object will only read in results of successful
## chains. Please use read_cmdstan_csv() to read the results of the failed chains
## separately. Use the $output(chain_id) method for more output of the failed
## chains.

## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 3.5 seconds.
## Chain 2 finished in 3.5 seconds.
## Chain 4 finished in 3.7 seconds.
## Chain 3 finished in 3.9 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 3.7 seconds.
## Total execution time: 4.1 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 3.2 seconds.
## Chain 4 finished in 3.2 seconds.
## Chain 3 finished in 3.3 seconds.
## Chain 2 finished in 3.5 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 3.3 seconds.
## Total execution time: 3.7 seconds.

```

```

##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 2.5 seconds.
## Chain 2 finished in 2.6 seconds.
## Chain 3 finished in 2.6 seconds.
## Chain 4 finished in 2.5 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 2.5 seconds.
## Total execution time: 2.8 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 4.3 seconds.
## Chain 4 finished in 4.2 seconds.
## Chain 2 finished in 4.4 seconds.
## Chain 3 finished in 4.5 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 4.4 seconds.
## Total execution time: 4.7 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 4 finished in 2.4 seconds.
## Chain 1 finished in 2.6 seconds.
## Chain 2 finished in 2.6 seconds.
## Chain 3 finished in 2.7 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 2.6 seconds.
## Total execution time: 3.0 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 3.6 seconds.
## Chain 3 finished in 3.7 seconds.
## Chain 2 finished in 3.8 seconds.
## Chain 4 finished in 3.9 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 3.7 seconds.
## Total execution time: 4.1 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 1.8 seconds.
## Chain 2 finished in 1.8 seconds.
## Chain 3 finished in 1.9 seconds.
## Chain 4 finished in 1.9 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 1.8 seconds.

```

```

## Total execution time: 2.1 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 3.8 seconds.
## Chain 2 finished in 3.8 seconds.
## Chain 4 finished in 3.9 seconds.
## Chain 3 finished in 4.2 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 3.9 seconds.
## Total execution time: 4.4 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 2 finished in 3.7 seconds.
## Chain 4 finished in 3.7 seconds.
## Chain 3 finished in 3.9 seconds.
## Chain 1 finished in 4.4 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 3.9 seconds.
## Total execution time: 4.6 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 2.5 seconds.
## Chain 3 finished in 2.5 seconds.
## Chain 2 finished in 2.5 seconds.
## Chain 4 finished in 2.5 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 2.5 seconds.
## Total execution time: 2.7 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 3.9 seconds.
## Chain 4 finished in 3.9 seconds.
## Chain 2 finished in 4.2 seconds.
## Chain 3 finished in 4.1 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 4.0 seconds.
## Total execution time: 4.4 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 4 finished in 3.8 seconds.
## Chain 2 finished in 3.9 seconds.
## Chain 1 finished in 4.0 seconds.
## Chain 3 finished in 4.0 seconds.
##
## All 4 chains finished successfully.

```

```

## Mean chain execution time: 3.9 seconds.
## Total execution time: 4.2 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 4 finished in 3.7 seconds.
## Chain 3 finished in 4.0 seconds.
## Chain 1 finished in 4.4 seconds.
## Chain 2 finished in 4.3 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 4.1 seconds.
## Total execution time: 4.5 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 3 finished in 2.4 seconds.
## Chain 1 finished in 2.5 seconds.
## Chain 4 finished in 2.4 seconds.
## Chain 2 finished in 2.8 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 2.5 seconds.
## Total execution time: 3.0 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 4.6 seconds.
## Chain 3 finished in 4.5 seconds.
## Chain 4 finished in 4.5 seconds.
## Chain 2 finished in 4.7 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 4.6 seconds.
## Total execution time: 4.8 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 3 finished in 3.8 seconds.
## Chain 1 finished in 3.9 seconds.
## Chain 2 finished in 4.0 seconds.
## Chain 4 finished in 4.2 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 4.0 seconds.
## Total execution time: 4.5 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 4 finished in 3.8 seconds.
## Chain 2 finished in 4.1 seconds.
## Chain 3 finished in 4.1 seconds.
## Chain 1 finished in 4.2 seconds.
##

```

```

## All 4 chains finished successfully.
## Mean chain execution time: 4.0 seconds.
## Total execution time: 4.4 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 3 finished in 4.4 seconds.
## Chain 4 finished in 4.6 seconds.
## Chain 1 finished in 4.8 seconds.
## Chain 2 finished in 4.8 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 4.7 seconds.
## Total execution time: 5.0 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 2 finished in 2.8 seconds.
## Chain 4 finished in 2.8 seconds.
## Chain 1 finished in 3.0 seconds.
## Chain 3 finished in 3.1 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 2.9 seconds.
## Total execution time: 3.3 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 3 finished in 5.0 seconds.
## Chain 1 finished in 5.2 seconds.
## Chain 2 finished in 5.2 seconds.
## Chain 4 finished in 5.2 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 5.1 seconds.
## Total execution time: 5.4 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 4.4 seconds.
## Chain 3 finished in 4.2 seconds.
## Chain 4 finished in 4.2 seconds.
## Chain 2 finished in 4.4 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 4.3 seconds.
## Total execution time: 4.6 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 2 finished in 4.0 seconds.
## Chain 1 finished in 4.2 seconds.
## Chain 3 finished in 4.3 seconds.
## Chain 4 finished in 4.3 seconds.

```

```
##
## All 4 chains finished successfully.
## Mean chain execution time: 4.2 seconds.
## Total execution time: 4.5 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 4.6 seconds.
## Chain 2 finished in 4.8 seconds.
## Chain 3 finished in 4.9 seconds.
## Chain 4 finished in 4.8 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 4.8 seconds.
## Total execution time: 5.0 seconds.
##
## Running MCMC with 4 chains, at most 14 in parallel...
##
## Chain 1 finished in 5.0 seconds.
## Chain 4 finished in 5.1 seconds.
## Chain 2 finished in 5.3 seconds.
## Chain 3 finished in 5.3 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 5.2 seconds.
## Total execution time: 5.6 seconds.
```

And we also fit a frequentist logistic regression model using the `glm` function from the `stats` package.

```
# run glm
fits_freq <- lapply(formulas, function(x)
  glm(x, family = binomial(link = "logit"), data = data))
```

Model Comparison

We get the log marginal likelihood with the `glm.logml.post` function from the `hdbayes` package and compute the AIC and BIC for each frequentist model.

```
# compute marginal likelihood
marginal_likelihoods <- lapply(fits_bayes, function(x) glm.logml.post(x)$logml)
# compute AIC
AICs <- lapply(fits_freq, function(x) x$aic)
# compute BIC
BICs <- lapply(fits_freq, function(x) BIC(x))

# create data frame
metrics <- data.frame(
  variables = unlist(covariates_models),
  logml = unlist(marginal_likelihoods),
  AIC = unlist(AICs),
  BIC = unlist(BICs)
)
```

```
# sort by logml
metrics <- metrics[order(metrics$logml, decreasing = TRUE), ]
```

```
head(metrics)
```

```
##  variables      logml      AIC      BIC
## 1      Ag -81.56298 146.7417 153.3383
## 3      D -84.53785 152.6359 159.2325
## 2      S -85.03840 152.4738 159.0705
## 4      Ch -85.38647 150.9365 157.5331
## 5      H -85.77183 157.8447 164.4414
## 6      W -86.17310 154.1318 160.7284
```

Weights

We compute the weights based on the log marginal likelihood, AIC, and BIC. For the log marginal likelihood, we use the `logSumExp` function from the `matrixStats` package to avoid numerical issues and compute the weights as $w_i = \exp(\log ml_i - \log \text{SumExp}(\log ml))$. For the AIC and BIC, we compute the weights as $w_i = \exp(-AIC_i - \log \text{SumExp}(-AIC))$ and $w_i = \exp(-BIC_i - \log \text{SumExp}(-BIC))$, respectively.

```
# compute weights
weights_logml <- exp(unlist(metrics$logml) - logSumExp(unlist(metrics$logml)))
weights_AIC <- exp(unlist(-metrics$AIC) - logSumExp(unlist(-metrics$AIC)))
weights_BIC <- exp(unlist(-metrics$BIC) - logSumExp(unlist(-metrics$BIC)))

# create a data frame with the weights
weights <- data.frame(
  variables = metrics$variables,
  logml = weights_logml,
  AIC = weights_AIC,
  BIC = weights_BIC
)
```

And we get the following weights ordered by log marginal likelihood:

Table 1: Weights based on log marginal likelihood, AIC, and BIC.

	variables	logml	AIC	BIC
1	Ag	0.8687128	0.0163642	0.6831463
2	D	0.0443511	0.0000451	0.0018823
3	S	0.0268856	0.0000530	0.0022135
4	Ch	0.0189825	0.0002467	0.0102976
5	H	0.0129121	0.0000002	0.0000103
6	W	0.0086442	0.0000101	0.0004217
7	Ag, H	0.0067880	0.0022231	0.0034288
8	Ag, W	0.0044156	0.1177606	0.1816260
9	Ag, D	0.0034101	0.0082869	0.0127811
10	Ag, S	0.0015711	0.0053151	0.0081977
11	Ag, Ch	0.0012317	0.0367278	0.0566466
12	D, H	0.0004126	0.0000079	0.0000122
13	H, W	0.0003848	0.0000395	0.0000610

14	S, H	0.0002508	0.0000086	0.0000132
15	Ch, H	0.0001711	0.0000422	0.0000651
16	D, Ch	0.0001656	0.0009198	0.0014186
17	Ch, W	0.0001322	0.0034587	0.0053345
18	S, D	0.0001171	0.0000123	0.0000190
19	S, Ch	0.0001058	0.0011026	0.0017005
20	D, W	0.0001006	0.0000433	0.0000667
21	S, W	0.0000832	0.0001331	0.0002053
22	Ag, H, W	0.0000710	0.0325655	0.0018556
23	Ag, D, H	0.0000260	0.0011303	0.0000644
24	Ag, S, H	0.0000122	0.0007257	0.0000414
25	Ag, D, W	0.0000112	0.0208674	0.0011891
26	Ag, Ch, H	0.0000095	0.0051125	0.0002913
27	Ag, Ch, W	0.0000075	0.4088858	0.0232991
28	Ag, S, D	0.0000069	0.0011253	0.0000641
29	Ag, S, W	0.0000064	0.0224222	0.0012777
30	Ch, H, W	0.0000042	0.0034559	0.0001969
31	Ag, D, Ch	0.0000039	0.0120994	0.0006894
32	D, H, W	0.0000024	0.0000323	0.0000018
33	S, H, W	0.0000022	0.0000922	0.0000053
34	Ag, S, Ch	0.0000020	0.0091222	0.0005198
35	D, Ch, H	0.0000016	0.0001345	0.0000077
36	S, D, H	0.0000011	0.0000020	0.0000001
37	S, Ch, H	0.0000009	0.0001554	0.0000089
38	D, Ch, W	0.0000005	0.0020426	0.0001164
39	S, Ch, W	0.0000005	0.0052182	0.0002973
40	S, D, Ch	0.0000004	0.0001991	0.0000113
41	S, D, W	0.0000003	0.0000188	0.0000011
42	Ag, D, H, W	0.0000002	0.0049812	0.0000105
43	Ag, Ch, H, W	0.0000001	0.0950341	0.0002001
44	Ag, S, H, W	0.0000001	0.0054693	0.0000115
45	Ag, S, D, H	0.0000001	0.0001535	0.0000003
46	Ag, D, Ch, H	0.0000000	0.0016928	0.0000036
47	Ag, S, D, W	0.0000000	0.0030650	0.0000065
48	Ag, D, Ch, W	0.0000000	0.0594803	0.0001252
49	Ag, S, Ch, H	0.0000000	0.0012800	0.0000027
50	D, Ch, H, W	0.0000000	0.0009629	0.0000020
51	S, Ch, H, W	0.0000000	0.0021465	0.0000045
52	Ag, S, Ch, W	0.0000000	0.0656575	0.0001382
53	S, D, H, W	0.0000000	0.0000125	0.0000000
54	Ag, S, D, Ch	0.0000000	0.0016458	0.0000035
55	S, D, Ch, H	0.0000000	0.0000282	0.0000001
56	S, D, Ch, W	0.0000000	0.0007091	0.0000015
57	Ag, S, D, H, W	0.0000000	0.0007404	0.0000001
58	Ag, D, Ch, H, W	0.0000000	0.0130737	0.0000010
59	Ag, S, Ch, H, W	0.0000000	0.0141904	0.0000011
60	Ag, S, D, Ch, H	0.0000000	0.0002304	0.0000000
61	S, D, Ch, H, W	0.0000000	0.0003026	0.0000000
62	Ag, S, D, Ch, W	0.0000000	0.0089513	0.0000007
63	Ag, S, D, Ch, H, W	0.0000000	0.0019842	0.0000000

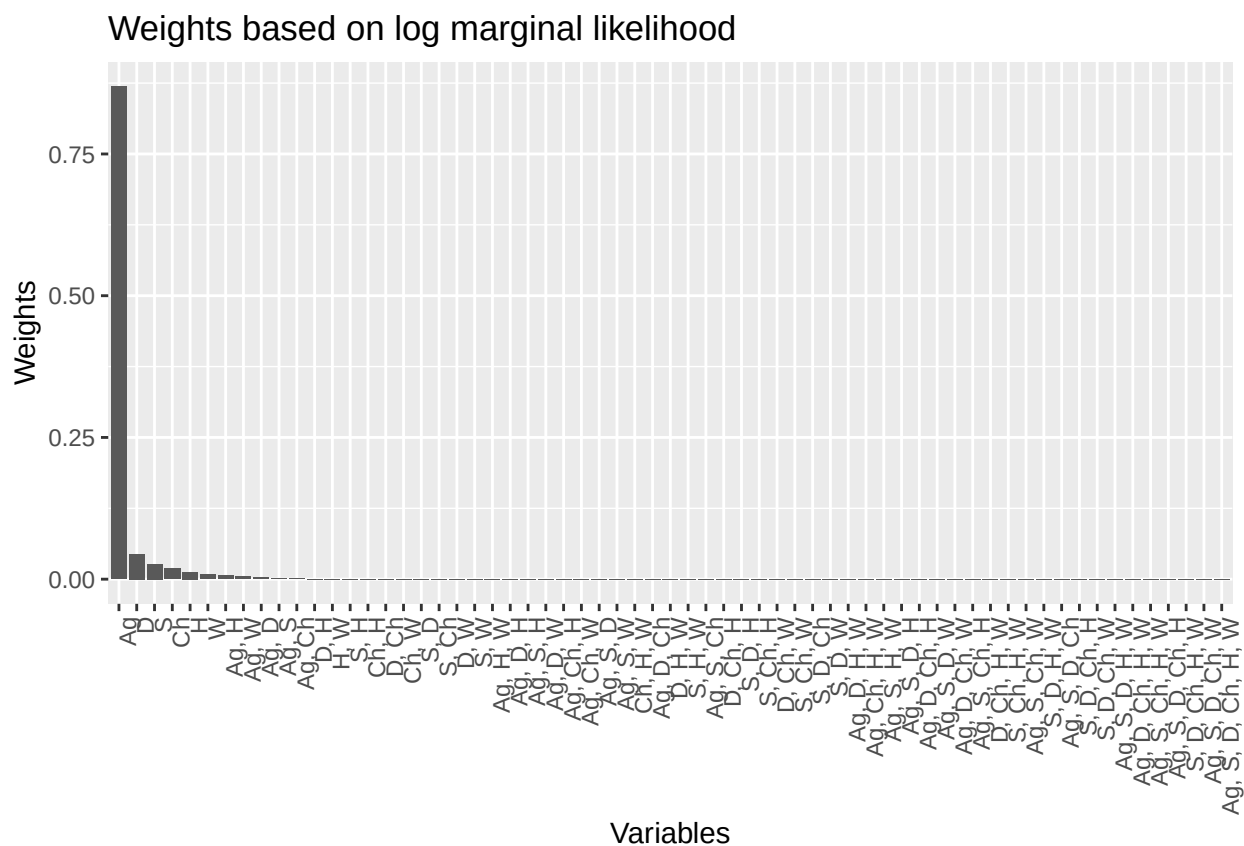
Then, best models based on log marginal likelihood are:

Table 2: Best weights based on log marginal likelihood.

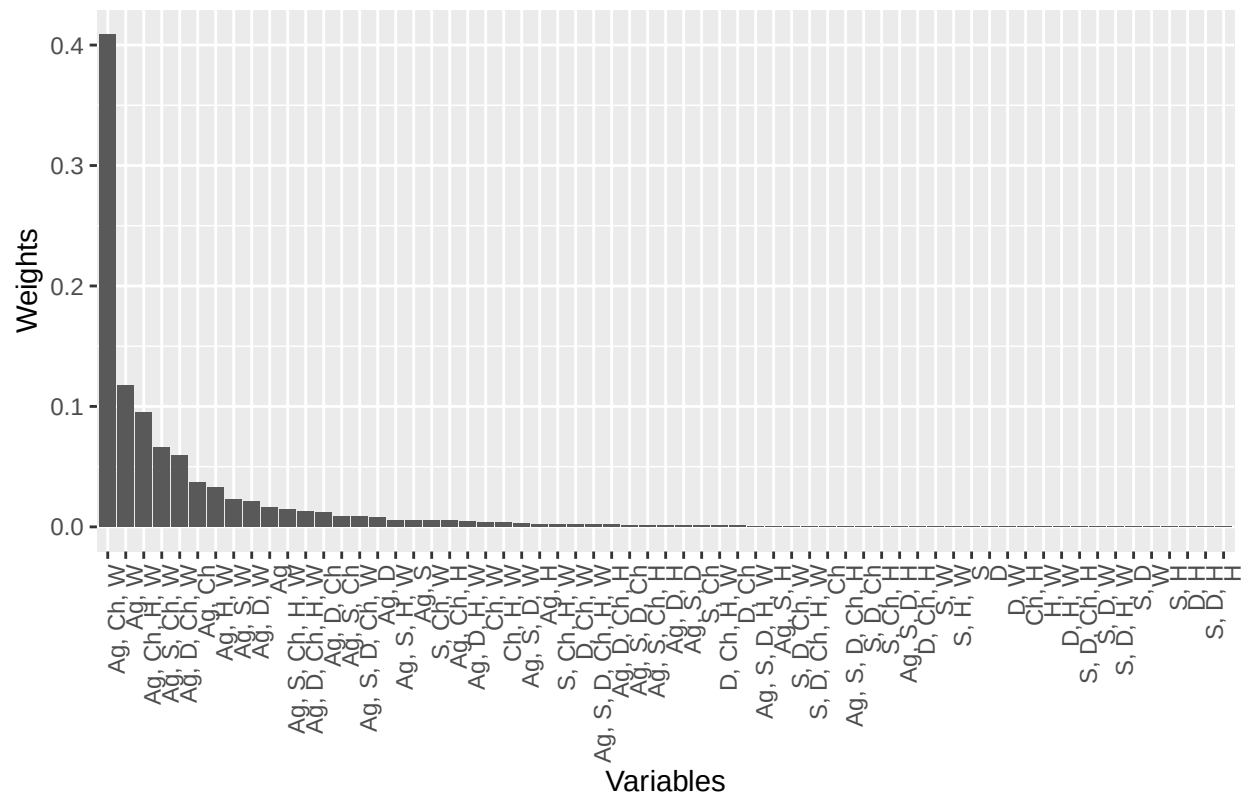
	variables	logml	AIC	BIC
1	Ag	0.8687128	0.0163642	0.6831463
2	D	0.0443511	0.0000451	0.0018823
3	S	0.0268856	0.0000530	0.0022135
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7	Ag, H	0.0067880	0.0022231	0.0034288
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9	Ag, D	0.0034101	0.0082869	0.0127811
10	Ag, S	0.0015711	0.0053151	0.0081977

Plots

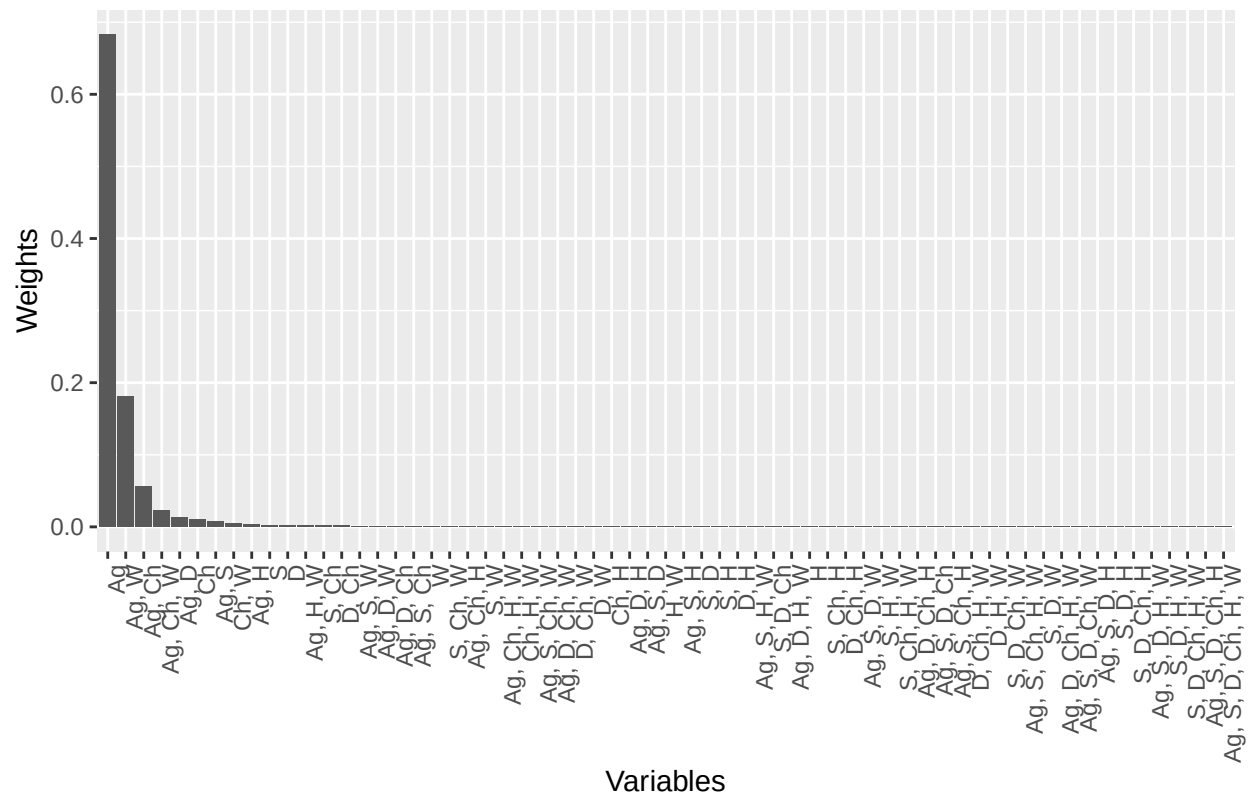
We plot the weights based on the log marginal likelihood, AIC, and BIC.



Weights based on AIC



Weights based on BIC



Log marginal likelihood vs BIC – Weights

